

Teacher-student dialogue (Topic: Stroop effect)

Teacher:

Student:

今天我们来介绍一个很有趣的心理学现象，叫做斯特鲁普效应。你听说过吗？

Today, we are going to introduce a very interesting psychological phenomenon called the Stroop effect. Have you heard of it?

嗯.....好像没怎么听过，能不能先简单说说它是什么？

Hmm... I don't think I've really heard of it. Could you briefly explain what it is first?

当然可以。斯特鲁普效应是指当我们看到一个词语的颜色和它的意思不一致时，比如‘蓝’字用红色印刷，我们的大脑会感到冲突，导致我们反应变慢或者容易出错。

Of course. The Stroop effect refers to the conflict our brain experiences when the color of a word is inconsistent with its meaning. For example, when the word "blue" is printed in red, our brain experiences interference, which makes us respond more slowly or more likely to make mistakes.

哦，意思是说，比如看到‘蓝’这个字是红色的，我会不自觉地想读‘蓝’，但又要说颜色‘红色’，这样就会有点混乱？

Oh, so it means that if I see the word "blue" printed in red, I will automatically want to read the word "blue," but I am supposed to say the color "red," which makes things a bit confusing?

对，就是这样。这个现象是由美国心理学家约翰·斯特鲁普在 1935 年通过实验发现的。

Yes, exactly. This phenomenon was discovered through experiments by the American psychologist John Stroop in 1935.

那他的实验是怎么做的呢？

How did he conduct the experiment?

他让参与者忽略词的意思，快速说出印刷的颜色。比如看到‘红’字是蓝色印刷，就要说‘蓝色’。结果发现，当词义和颜色冲突时，反应时间平均延长了大约 74 毫秒，而且错误率也更高。

He asked participants to ignore the meaning of the word and quickly name the color in which it was printed. For example, if the word “red” was printed in blue, they had to say “blue.” The results showed that when the word meaning and color conflicted, response time increased by about 74 milliseconds on average, and the error rate was also higher.

74 毫秒听起来不多，但我猜对心理学来说，这很明显吧？

Seventy-four milliseconds does not sound like much, but I guess in psychology, that is quite noticeable, right?

没错，虽然时间很短，但足以说明认知上的冲突。因为阅读词义是我们大脑自动化的过程，而颜色识别需要更多的注意力控制。

Exactly. Although the delay is very short, it is enough to indicate cognitive conflict. Reading word meanings is an automatic process for the brain, whereas identifying colors requires more attentional control.

自动化和受控加工？这两个概念能详细说说吗？

Automatic and controlled processing? Could you explain these two concepts in more detail?

好的。自动化加工指的是我们不费力、几乎无意识完成的任务，比如阅读词义。而受控加工则需要有意识地集中注意力，比如判断颜色。冲突出现时，自动化加工会抢先占用注意资源，干扰颜色判断。

Sure. Automatic processing refers to tasks that we complete with little effort and almost unconsciously, such as reading word meanings. Controlled processing, on the other hand, requires consciously focusing attention, such as identifying colors. When conflict occurs, automatic processing takes up attentional resources first and interferes with color judgment.

哦，那就是说我们大脑习惯先读词义，反而影响了颜色识别？

Oh, so our brain is used to reading the word meaning first, and that ends up interfering with color recognition?

完全正确。为了正确回答颜色，我们需要抑制自动阅读的冲动，这会消耗额外的认知资源，导致反应变慢。

Absolutely. To correctly name the color, we need to suppress the automatic impulse to read the word, which consumes additional cognitive resources and slows down our response.

这有点像我玩游戏时，看到颜色和提示不符，会迟钝一下。

This is a bit like when I play a game and the color does not match the prompt, which makes me hesitate for a moment.

很好的比喻！其实后来研究还发现了很多变体，比如情绪斯特鲁普任务，负面词汇更容易吸引注意力，干扰更大。

That is a great analogy. Later studies also identified many variants, such as the emotional Stroop task, in which negative words are more likely to capture attention and cause stronger interference.

情绪词也会影响？那如果写‘恐惧’这个词用红色印刷，会更难回答颜色吗？

Emotional words can also have an effect? So if the word “fear” is printed in red, would it be harder to name the color?

对，负面词会更强烈地抓住我们的注意力，导致颜色命名更慢。还有数字斯特鲁普任务，比如数字的数量和数值不一致，也会引发类似冲突。

Yes. Negative words capture our attention more strongly, making color naming slower. There is also the numerical Stroop task, where conflicts between the quantity of digits and their numerical value can produce similar interference.

哇，这个效应应用范围还挺广的嘛。那它会不会受年龄或者情绪影响？

Wow, this effect seems to have quite a wide range of applications. Can it be influenced by age or emotional state?

是的，年龄、情绪状态、任务难度和刺激呈现时间都会影响斯特鲁普效应的大小。

Yes. Age, emotional state, task difficulty, and stimulus presentation time can all affect the magnitude of the Stroop effect.

那有没有办法训练，减少这种干扰？

Is there any way to train people to reduce this interference?

有的。通过系统训练，人们可以增强执行注意力控制能力，从而减弱斯特鲁普效应的影响。

Yes. Through systematic training, people can strengthen their executive attentional control and thereby reduce the influence of the Stroop effect.

听起来对生活挺有帮助的。那这个效应在设计方面有什么启示吗？

That sounds quite useful in everyday life. What implications does this effect have for design?

当然。它告诉我们，在界面设计中，文字、颜色和图标信息最好保持一致，避免认知冲突。比如一个绿色的‘警告’标签，就可能引发斯特鲁普效应，增加误操作风险。

Of course. It tells us that in interface design, text, color, and icon information should ideally be consistent to avoid cognitive conflict. For example, a green “warning” label may trigger a Stroop-like conflict and increase the risk of user errors.

对啊，我以前用过某个软件，按钮颜色和功能不匹配，老是点错。原来是因为这个原因。

That makes sense. I once used a piece of software where the button colors did not match their functions, and I kept clicking the wrong buttons. So this may be the reason.

正是如此。医疗领域也用斯特鲁普任务来评估认知衰退，通过分析反应时间和手部动作。

Exactly. In the medical field, Stroop tasks are also used to assess cognitive decline by analyzing response time and hand movements.

那智能体训练里也用吗？比如机器人学习同时优化速度和精度？

Is it also used in agent training, such as teaching robots to optimize both speed and accuracy?

对，强化学习中引入类似斯特鲁普任务的多目标冲突情境，可以训练智能体学会监测内部冲突，灵活分配资源。

Yes. Introducing multi-objective conflict scenarios similar to Stroop tasks into reinforcement learning can train agents to monitor internal conflict and flexibly allocate resources.

这真是太酷了！原来斯特鲁普效应不仅是心理学实验，还能应用到这么多领域。

That is really cool. It turns out that the Stroop effect is not only a psychological experiment, but can also be applied in so many fields.

没错，它帮助我们理解认知控制和注意力调节，是连接理论和实际的桥梁。

Exactly. It helps us understand cognitive control and attentional regulation, serving as a bridge between theory and practice.

谢谢老师，今天学到了好多，感觉大脑的工作原理真神奇。

Thank you, teacher. I learned a lot today, and the way the brain works feels truly fascinating.

很高兴你喜欢！下次我们可以再学习其他设计心理学现象。

I am glad you enjoyed it. Next time, we can learn about other psychological phenomena related to design.